

Hurghada Hospital Management System (HHMS)

Project Proposal

Course: Software Engineering Instructor: Dr. Wesam Ahmed Date: October 11, 2026

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1. The Problem

Hospitals in Hurghada serve local residents and a large number of tourists. When appointments, patient records, prescriptions and pharmacy stock are handled on paper or in separate systems, patients wait longer, doctors do not see a patient's full history, medicines are prescribed without knowing what is in stock, and sensitive medical data is hard to protect.

2. Our Solution

HHMS is a web application that covers the whole patient journey: registration and appointment booking, the doctor's diagnosis and prescription, pharmacy dispensing and payment. It works as a secure electronic medical record (EMR) for the hospital staff and gives patients an online portal to manage their own appointments and view their records.

3. Intended Users and How They Use the System

User	How they use HHMS
Patient	Signs up online with an Egyptian national ID or, for tourists, a passport number. Books and cancels appointments, views their own medical history, prescriptions and receipts, and receives appointment reminders.
Receptionist	Registers walk-in patients, books and cancels appointments on their behalf, checks patients in on arrival and records appointment payments. Sees contact details only, never medical data.
Doctor	Sees the day's schedule, opens the medical history of their own patients, records the complaint, diagnosis and notes of each visit, and writes prescriptions.
Pharmacist	Works from a queue of prescriptions waiting to be filled, dispenses medicines (stock is updated automatically) and manages stock, with an alert when a medicine runs low.
Admin	Manages staff accounts, departments and their consultation fees, and each doctor's weekly working hours.

Example: a tourist with a fever signs up with a passport number and books a 10:30 slot in Internal Medicine. At the hospital, the receptionist checks the patient in and records the consultation fee. The doctor checks the patient's allergies, records the diagnosis and prescribes two medicines. The pharmacist sees the prescription in the queue and dispenses it, and the stock is updated. Afterwards, the patient can see the visit, the prescription and the receipt online.

4. Overall Functionality

- **Accounts and access control:** secure login with five roles; each role sees only what it needs, and doctors see only their own patients.
- **Patient records:** online or walk-in registration and a medical history made of the patient's background (allergies, chronic diseases, blood type, past surgeries, family history) and a record of every visit.
- **Doctors and departments:** departments with consultation fees, doctors and their weekly working hours.
- **Appointments:** booking free slots within a doctor's working hours, cancellation, check-in and the doctor's daily schedule.
- **Prescriptions:** one or more medicines per visit with dosage and quantity, visible to the patient and to the pharmacy.
- **Pharmacy inventory:** dispensing linked to stock levels, with low-stock alerts.
- **Billing:** appointment payments recorded at reception, with receipts the patient can view.

- **Automated notifications:** appointment reminders, booking and cancellation confirmations, and low-stock alerts.

5. How HHMS Helps Its Users

- Patients book from anywhere and keep their whole medical history in one place.
- Doctors see the complete history before diagnosing, including allergies before prescribing.
- Reception avoids double bookings and paper receipts.
- The pharmacy always knows what is in stock and what is waiting to be dispensed.
- Medical data stays private through role-based access, and two staff members editing the same record at the same time cannot overwrite each other's changes.

6. Scope and Technology

We are building HHMS with ASP.NET Core (C#), SQL Server, Next.js and n8n for automations. Hospital admission is outside the current scope and is planned as future work.